

6.8 DIFFERENTIAL EQUATIONS, DE (4758) A2

Objectives

To extend the work in *Applications of Advanced Mathematics (C4)* on the formulation and solution of differential equations. Students are expected to have a reasonable degree of manipulative competence and to be able to handle more complicated problems.

Assessment

Examination (72 marks)
1 hour 30 minutes
Candidates answer three questions from four, each worth 24 marks.

Coursework (18 marks)
One modelling assignment involving the use of differential equations at an appropriate level of sophistication.
There are no restrictions on the context chosen.

Unless otherwise specified the value of the acceleration due to gravity should be taken to be exactly 9.8 ms^{-2} .

Assumed Knowledge

Candidates are expected to know the content of *C1*, *C2*, *C3* and *C4*.
In addition candidates are expected to know basic kinematics and Newton's Second Law.
Relevant knowledge of complex numbers will also be required.

Calculators

In the MEI Structured Mathematics specification, no calculator is allowed in the examination for *C1*.
For all other units, including this one, a graphical calculator is allowed.

DIFFERENTIAL EQUATIONS, DE		
Specification	Ref.	Competence Statements

MODELLING WITH DIFFERENTIAL EQUATIONS

Constructions of models.	DEp1	Understand how to introduce and define variables to describe a given situation in mathematical terms.
	2	Be able to relate 1 st and 2 nd order derivatives to verbal descriptions and so formulate differential equations.
	3	Know the language of kinematics, and the relationships between the various terms.
	4	Know Newton's 2 nd law of motion.
	5	Understand how to determine the order of a differential equation.
Interpretation of solutions.	6	Be able to interpret the solution of a differential equation in terms of the original situation.
	7	Appreciate the difference between a general solution and a particular solution, i.e. one which satisfies particular prescribed conditions.
	8	Understand the significance of the number of arbitrary constants in a general solution.
	9	Be able to investigate the effect of changing a differential equation on its solution.
Tangent fields.	10	Be able to sketch the tangent field for a 1 st order differential equation and be able to interpret it.
	11	Be able to sketch and interpret the curve of the solution corresponding to particular conditions.
	12	Be able to identify isoclines and use them in sketching and interpreting tangent fields.

FIRST ORDER DIFFERENTIAL EQUATIONS

Equations with separable variables.	DEc1	Be able to find both general and particular solutions of a 1 st order differential equation with separable variables.
First order linear differential equations.	2	Be able to solve 1 st order linear differential equations with constant coefficients.
	3	Be able to distinguish differential equations where the integrating factor method is appropriate, and to rearrange such equations if necessary.
	4	Be able to find an integrating factor and understand its significance in the solution of an equation.
	5	Be able to solve an equation using an integrating factor and find both general and particular solutions.

DIFFERENTIAL EQUATIONS, DE			
Ref.	Notes	Notation	Exclusions

MODELLING WITH DIFFERENTIAL EQUATIONS
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DEp1	Candidates will be expected to model real-life situations with differential equations.	The rate of change of x with respect to time $= \dot{x}$ (etc.)	
2	The differential equations will not be restricted to those which candidates can solve analytically.		
3	Including acceleration $= v \frac{dv}{ds}$.	$v = \frac{ds}{dt}$ $a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$	
4	In the form $F = ma$.		Variable mass.
5			
6			
7			
8			
9			
10	e.g. changing the values of coefficients.		
11		The term direction field has the same meaning as tangent field.	
12	e.g. the solution through a particular point.		

FIRST ORDER DIFFERENTIAL EQUATIONS

DEc1			
2	Equations of the form $y' + ay = 0$ and $y' + ay = f(x)$ in simple cases, where a is constant.		
3	Equations which can be expressed in the form: $\frac{dy}{dx} + P(x)y = Q(x)$.		
4	Integrating factor $e^{\int P dx}$.		
5	e.g. through a particular point.		

DIFFERENTIAL EQUATIONS, DE		
Specification	Ref.	Competence Statements

SECOND AND HIGHER ORDER DIFFERENTIAL EQUATIONS

Homogeneous second order linear differential equations.	DEc6	Be able to solve homogenous 2 nd order differential equations, using the auxiliary equation and complementary function.
	7	Appreciate the relationship between different cases of the solution and the nature of the roots of the auxiliary equation, and be able to interpret these different cases graphically.
	8	Be able to find the particular solution in given contexts.
The general second order linear differential equation.	9	Be able to solve the general 2 nd order linear differential equation, by solving the homogeneous case and adding a particular integral.
	10	Be able to find particular integrals in simple cases. Appreciate the relationship between different cases of the solution and the nature of the roots of the auxiliary equation, and be able to interpret these different cases graphically.
	11	Be able to solve the equation for simple harmonic motion, $\ddot{x} + \omega(x + k) = 0$, and be able to relate the various forms of the solution to each other.
Damped oscillations.	12	Be able to model damped oscillations using 2 nd order linear differential equations, and understand the associated terminology.
	13	Be able to interpret the solutions of equations modelling damped oscillations in words and graphically.
Higher order linear differential equations.	14	Appreciate that the same methods can be extended to higher order equations and be able to do so in simple cases.

SIMULTANEOUS DIFFERENTIAL EQUATIONS
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Simultaneous linear differential equations.	DEe1	Model situations with one independent variable and two dependent variables which lead to 1st order simultaneous differential equations, and know how to solve these by eliminating one variable to produce a single, 2 nd order equation.
	2	Appreciate that the same method can be extended to more than two such equations, leading by elimination to a single higher order equation.

NUMERICAL METHODS

Step by step methods.	DEs1	Be able to use step by step methods (e.g. Euler's method) to solve 1 st order differential equations (including simultaneous equations) where appropriate.
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DIFFERENTIAL EQUATIONS, DE			
Ref.	Notes	Notation	Exclusions

SECOND AND HIGHER ORDER DIFFERENTIAL EQUATIONS

DEc6	Equations of the form $y'' + ay' + by = 0$ where a and b are constants.		
7	Discriminant $> 0, = 0, < 0$. A basic understanding of the complex roots of a quadratic equation is assumed.		
8			
9	Equations of the form $y'' + ay' + by = f(x)$ where a and b are constants.		
10	Cases where $f(x)$ is a polynomial, trigonometric or exponential function.		
11			
12			
13	The damping will be described as 'over-', 'critical' or 'under-' according to whether the roots of the auxiliary equation are real distinct, equal or complex.		
14	Adequate guidance will be given in examination questions.		

SIMULTANEOUS DIFFERENTIAL EQUATIONS
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DEe1	Applications such as predator-prey models.		
2	Adequate guidance will be given in any examination question involving more than two dependent variables.		

NUMERICAL METHODS

DEs1	Adequate guidance will be given in examination questions.	Specific step-by-step algorithms.
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Differential Equations (DE) Coursework: How Differential Equations are used to Solve Real-World Problems

Rationale

The aims of the coursework are that students should learn how differential equations are used to solve real-world problems and that they should appreciate how the theory they have learnt for the examination helps them to do this.

The objectives are that they should be able to undertake the various steps in the problem solving procedure shown in the flow chart in Section 5.2. The assessment criteria are closely related to these steps.

Description

There are two aspects to the work.

- (i) The modelling cycle consists of pen and paper development of the consequences of the basic assumptions made, leading to a predicted outcome which must then be tested against reality.
- (ii) In the experimental cycle, results are collected in order to give insight into the situation under investigation, so that a realistic model can be developed.

Level of Work

The task represents 20% of the assessment and the work involved should be consistent with that figure, both in quantity and level of sophistication. Tasks which allow only superficial or trivial treatment should be avoided.

Assessment

Each task must be assessed on one of the coursework assessment sheets, A or B. The assessor decides on the appropriate sheet according to the way the candidate has approached the particular task.

- (A) In this case the modelling cycle is investigated in some depth, whilst the check against reality may use the data from published sources, from experiments which the candidate has not actually performed or from experience; there must however be a quantitative element in such data.
- (B) The work presented is approximately evenly divided between developing the model, and one or more experiments conducted by the candidate to verify the quality of predictions from it and/or to inform its development.

No other mark sheet may be used, nor may these be amended in any way.

One mark is available for each criterion statement. Half marks may be awarded, but the overall total must be rounded (up or down) to an integer.

Note that in the case of Mark Scheme A, the marks for ‘Manipulating the Model’ may be awarded for the quality of the work either on the first or the second modelling cycles.

Task Selection

Centres are encouraged to develop their own coursework tasks. If they have any doubt about the suitability of a proposed task, they are recommended to submit details of it to the Principal Coursework Moderator, via OCR.

However, Centres which are new to the scheme are strongly recommended to use the tasks published by MEI while they are familiarising themselves with the nature of coursework. They should ensure that the material they have is that published for this specification and not that for an earlier specification.

Centres are advised that the choice of suitable tasks is crucial to the success of their candidates’ coursework.

Differential Equations (DE) Coursework: Assessment Sheet A (Mark Scheme A) Work Based on the Modelling Cycle

Task: Candidates will model a real-life situation of their own choice which requires the use of differential equations.

Coursework Title												
Candidate Name							Candidate Number					
Centre Number							Date					
Domain	Mark	Description						Comment	Mark			
Simplifying the situation and setting up the model.	1	There are clear statements of the assumptions made and an awareness is shown of the relative importance of these assumptions.										
	1	The relevance of the assumptions to the initial equations of the model is explained clearly.										
	1	Differential equation(s) to model the situation are established and justified.										
Manipulating the model.	1	A correct method of solution is applied to the differential equation(s).										
	1	A solution to the differential equation(s) is obtained.										
	1	Values for the parameters of the equation(s) are chosen and justified.										
	1	There is a quantitative consideration of the effects of possible variation in the values of the parameters.										
The collection of data to verify the model.	1	The source and means of collecting the data are clearly described and their relevance to the model is demonstrated.										
	1	The data are presented in a form suitable for comparison with the predictions of the model.										
Comparison between the data collected and the predictions of the model.	1	A clear comparison is made, using diagrams and graphs where appropriate.										
	1	The effects of variation in the parameters are taken into account.										
Revision of the model.	1	In the light of the comparison, proposals are made for amending the initial assumptions to improve the quality of the model.										
	1	New equations are established as a result of the amended assumptions.										
Assessment of the improvement obtained.	1	The new equations are manipulated to produce new predictions.										
	1	A comparison is made between the data and the new predictions, and comments made on whether an improvement has been obtained.										
Oral communication	2	Presentation		Please tick at least one box and give a brief report								
		Interview										
		Discussion										
Half marks may be awarded but the overall total must be an integer. Please report overleaf on any help that the candidate has received beyond the guidelines								TOTAL	18			

Coursework must be available for moderation by OCR

Differential Equations Coursework: Assessment Sheet B (Mark Scheme B)

Work Based On A Combination Of The Modelling And Experimental Cycles

Task: Candidates will model a real-life situation of their own choice which requires the use of differential equations.

Coursework Title												
Candidate Name							Candidate Number					
Centre Number							Date					
Domain	Mark	Description						Comment		Mark		
Simplifying the situation and setting up the model.	1	There are clear statements of the assumptions made and an awareness is shown of the relative importance of these assumptions.										
	1	The relevance of the assumptions to the initial equations of the model is explained clearly.										
	1	Differential equation(s) to model the situation are established and justified.										
Manipulating the model.	1	A correct method of solution is applied to the differential equation(s).										
	1	A solution to the differential equation(s) is obtained.										
	1	Values for the parameters of the equation(s) are chosen and justified.										
	1	A set of predictions is produced from the equation(s).										
Conducting the experiment.	1	The relevance of the assumptions to the design of the experiment is explained clearly.										
	1	The conduct of the experiment is designed clearly, including diagrams of the apparatus used. Any steps taken to reduce experimental error are also described.										
	1	Sufficient results are obtained, and these are presented clearly and concisely, in a form suitable for comparison with the predictions of the model.										
	1	There is a discussion of the variability in the measurements taken.										
Comparison between the experimental results and the predictions of the model.	1	A clear comparison is made, using diagrams and graphs where appropriate.										
	1	The effects of variation in the parameters and in the measurements are taken into account.										
Revision of the process.	1	In the light of the comparison, a decision is made, and justified, on whether further revision is needed to the modelling process, or to the conduct of the experiment, or both.										
	1	If revision is needed (or if the original assumptions led to a trivial piece of modelling), detailed proposals are made for amending the assumptions and/or the conduct of the experiment. If not, there is a serious discussion of how the work could be extended to related situations.										
Assessment of the improvement obtained.	1	There is an investigation into applying the effects of the proposed revision/extension. [Note: It is not necessary for the candidate to rework the model completely unless the initial differential equation(s) was(were) trivial.]										
Oral communication	2	Presentation		Please tick at least one box and give a brief report								
		Interview										
		Discussion										
Half marks may be awarded but the overall total must be an integer. Please report overleaf on any help that the candidate has received beyond the guidelines								TOTAL		18		

Coursework must be available for moderation by OCR